

AI on a one-person sports-cards store.

An honest inventory of every LLM call, every rules-based agent, every data source, and the multi-agent Claude Code workflow that runs it all. Built by Jason Chletsos. 2026-05-26.

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PRICING SOURCES

TL;DR. Three production LLM surfaces (all Anthropic Claude), each with a sharp job — buyer-facing chat assistant, per-listing copy/CTR optimizer, and on-phone card vision. Around the LLMs sits a deterministic agent constellation that does the slow, dollarized work: repricing, promoted-listings bidding, best-offer auto-respond, photo audits, Cassini scoring. Every production write is gated by a dry-run + typed-yes confirm. A custom Fivetran Connector SDK pipes five eBay tables into the warehouse; the buyer-facing site reads from that. Day-to-day development is driven by a multi-agent Claude Code cadence — fan out 4-6 specialist agents in parallel, verify, ship.

I. Live LLM surfaces

Three Anthropic Claude surfaces are in production today. Zero OpenAI, Gemini, or non-Anthropic calls in the codebase.

SURFACE	MODEL	WHAT IT DOES	COST & SAFETY
Buyer-site AI assistant LIVE	claude-opus-4-7	Lambda route <code>POST /ebay/ai-chat</code> backs <code>docs/assistant.html</code> . System prompt frames Claude as the store's cards assistant, with persona context (favorite Pokemon list, store policies). Multi-turn conversation; serves visitors who want help finding cards or asking buying questions.	max_tokens 1024; per-turn history truncated to last N, each capped at 8000 chars; token usage logged from response.
AI Listing Optimizer LIVE	claude-sonnet-4	Per-listing "AI Optimize" button on the seller dashboard. Sends current title, price, CTR, impressions, and promoted flag. Returns a 4-section diagnosis (DIAGNOSIS, OPTIMIZED TITLE, OPTIMIZED DESCRIPTION, ACTIONS). Streams via SSE so the diagnosis renders as it generates.	max_tokens 1200; BYO key (stored in browser localStorage); <code>anthropic-dangerous-direct-browser-access: true</code> .

SURFACE	MODEL	WHAT IT DOES	COST & SAFETY
Mobile card scanner (vision) PAUSED	claude-sonnet-4-6 (default) / opus-4-7 / haiku-4-5	Front + optional back photo, base64-encoded to the Messages API. System prompt forces a strict JSON schema — name, set, number, total, rarity, foil, edition, language, condition_hints, confidence. Output drops into the same <code>inventory.csv</code> schema the Python pipeline consumes, so card-scanner and CSV-import are interchangeable inputs.	max_tokens 600; API key in iOS Keychain / Android encrypted prefs via <code>expo-secure-store</code> ; per-scan cost surfaced in the model picker (~\$0.005 Sonnet, ~\$0.002 Haiku, ~\$0.03 Opus).

2. Heuristic agent stack (no LLM, by design)

Around the LLM surfaces sits a deterministic agent constellation. Each agent runs in **dry-run mode by default**, writes a plan JSON, renders an HTML report for review, and only mutates eBay state when invoked with `--apply` and an explicit per-agent confirmation. Total: 21 agents.

Pricing & offers

- **repricing_agent** — guardrailed raise/lower/hold from sold-history median, multi-source pricing, per-listing locks. Writes via Trading `ReviseItem`.
- **best_offer_agent** / **best_offer_autorespond_agent** — auto-accept at market median * 0.95, auto-decline at * 0.75, midpoint counter in the middle band.
- **watchers_offer_agent** — proactive offers to watchers using floor = max(absolute, current * multiplier, sold_median * 0.92).
- **price_drops_agent, price_consistency_agent** — flag listings off-market vs. comps.
- **card_price_agent** — tokenizes titles, scores candidate products against SportsCardsPro `/api/products`, fetches grade-matched price.

Listing performance & ranking

- **cassini_score_agent** — 0-100 rubric (photos 25, specifics 20, title 15, impressions 15, CTR 10, recent sale 10, offer eligibility 5).
- **listing_performance_agent** — reads eBay Sell Analytics `traffic_report` joined to local snapshot.
- **promoted_listings_agent** — 5 ad-rate tiers (NO_AD, LOW, STANDARD, AGGRESSIVE, MAX)

Inventory + ops

- **inventory_agent** — reads `inventory.csv` (now CollX-sourced), suggests title/category/price/specifics, renders ready-to-list HTML.
- **collx_ingest** — adapter from CollX Pro CSV export to the inventory schema (CollX has no public dev API).
- **push_to_ebay** — single-card live `AddItem` with dry-run + typed-yes confirm gate.
- **relist_agent** — ended/unsold republication queue.
- **orders_watch_agent, tracking_responder_agent** — order lifecycle + automated tracking replies.
- **combined_shipping** — flips `ApplyShippingDiscount` in bulk via `ReviseItem`.

Buyer-side & marketing

- **buyer_watchlist_agent** — scoring + outreach signal for watcher cohorts.
- **repeat_buyers_agent** — tier by count + lifetime spend; templated thank-yous.
- **email_campaign_agent** — assembles top-margin "Steals" plus volume-discount banner.
- **message_responder_agent** — FAQ pattern matching for buyer questions.

by rule classification.

- **photo_audit_agent / photo_quality_audit** — Pillow-based scoring on count, long-edge px, file-size proxy, aspect, coverage.

- **daily_digest_agent, pnl_agent, pokemon_news_agent, pokemon_deals_agent, top_sellers_agent, under_10_agent, vault_eligibility, hub_pages_agent** — reporting / merchandising / catalog.

3. Data + ML inputs

The agents are only as good as the signals they ingest. Five integration paths feed them:

- **Fivetran custom connector** (`connector.py`) — written against the Fivetran Connector SDK. Syncs five eBay tables to the warehouse: `active_listings`, `listing_performance`, `orders`, `promoted_listings`, `seller_standards`. Encodes hard-won knowledge — Browse API returns 403 for user tokens, so listing data is reconstructed from orders + promoted listings; `traffic_report` requires `marketplace_ids:EBAY_US`.
- **Pricing** — SportsCardsPro / PriceCharting (paid, single API key) as "actual price"; eBay Finding API sold history as "market price"; Pokemon TCG API (free) as Pokemon fallback; CollX Pro CSV as live market signal for current inventory.
- **Cassini ranking model** — deterministic 0-100 score derived from observable Cassini inputs. Not a trained model; an opinionated proxy.
- **Warehouse** — DuckDB locally, Fivetran-loaded `tester.*` tables consumed by the dashboard builder.
- **Lambda layer** — `jw0hur2091.execute-api.us-east-1.amazonaws.com` hosts the Trading API write surfaces (revise / reprice / ai-chat / oauth callback / account-deletion notification handler).

4. Collaboration workflow with Claude Code

The most active AI surface on this project is actually the development loop itself — Claude Code as a teammate, not as a generation tool. Three patterns are durable:

- **Multi-agent fan-out for features.** When a task is broad ("audit the buyer site for conversion gaps", "build the store-companion mobile flow"), launch 4–6 specialist agents in parallel — reseller persona, mobile-UX designer, eBay API risk reviewer, etc. Aggregate findings, ship. Default cadence.
- **Dry-run + typed-yes for every production write.** No agent ever writes to eBay without printing the full plan, surfacing economics + risks, and prompting for an explicit "yes". Encoded in `harpua-daily` skill, in `push_to_ebay.py`, in every `--apply` flag.
- **Persistent project memory.** Ten memory notes at `.claude/projects/.../memory/` capture project goal, agent catalog, conversion-audit findings, slang preferences, fan-out cadence. Future sessions start with full context — no "what are we working on again."

One concrete skill — `harpua-daily` — runs all 21 planning agents in dry-run, summarizes each into a single table (Agent · What it would do · \$ impact), and refuses to add `--apply` without explicit per-agent confirmation. The skill also encodes a known schema-drift bug in two agents and the one-line fix.

5. Paused / experimental

- **Mobile companion app** (Expo / React Native). Card scanner is wired to Claude vision; eBay listing creation is implemented. Paused at commit `2fbd691` — CollX Pro became the inventory source-of-truth and supplanted the on-phone scanner. Code is preserved, not deleted.
- **eBay Vault enrollment** — flagged by `best_offer_agent` but eBay has not exposed a Vault-enroll field in `ReviseItem` as of mid-2026. Will wire when the API surface lands.
- **Photo upload to live listings** — UI built; Lambda route `/upload-photos` not deployed.
- **Two known-broken agents** — `listing_performance_agent` , `pnl_agent` crash on snapshot schema drift. `harpua-daily` skips them; one-line fix is documented in the skill.

Generated by `build_ai_overview.py` from the live repo state. Source paths and findings traceable in the audit log.

harpua2001 store: ebay.com/str/harpua2001 · Public dashboard: [index.html](#) · Inventory pipeline: [inventory.html](#)